Working with BigQuery resources

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Overview

This notebook provides examples of working with data from a referenced resource.
Build upon the best practices demonstrated in this notebook to visualize and analyze data in your own workspaces.

Objective

Use this notebook to perform common workspace resource operations including:

- Access data from referenced resources
  - Read data from BigQuery
    - Using cell magics
    - Using pandas-gbq library

How to run this notebook

Run the Notebook setup section before running the cells of the other sections.

Costs

This notebook takes less than a minute to run, which will typically cost less than $0.01 of compute time on your cloud environment.

Notebook setup

Run the cell below to import dependencies and utilities.

Workspace setup

> **Note:** This notebook assumes that `workspace_setup.ipynb` has been run.

`workspace_setup.ipynb` creates two Cloud Storage buckets for your workspace files with workspace reference names:

- ws_files
- ws_files_autodelete_after_two_weeks

The code in this notebook will write output files to the "autodelete" bucket by default.
Any file in this bucket will be automatically deleted **two weeks** after it is written.

This alleviates the need for you to remember to clean up temporary and example files manually. If you want to write outputs to a durable location, simply change the assignment of the `BUCKET_REFERENCE` variable in the cell below and re-run the notebook.

Working with data from referenced resources

Referenced resources represent data or other elements in Verily Workbench by pointing to a source that exists outside of the current workspace. To add a referenced resource to your workspace, use the working_with_resources.ipynb notebook.

Reading data from BigQuery

There are many ways to interact with your referenced resource data in a Verily Workbench cloud environment. This notebook provides examples of several options, each of which are appropriate for certain use cases.

Using cell magics

For troubleshooting, demos and developing new queries, it's useful to leverage the IPython magics for BigQuery, several examples of which are provided below.

> **Note:** The examples in this sub-section utilize data from the 1000 Genomes data collection. Please add the BigQuery tables from that data collection to your current workspace before running the cells below. Instructions for adding a data collection to your workspace can be found here.

Getting summary stats

To get summary statistics and visualizations for all the columns of a BigQuery table, you can use the `%bigquery_stats` cell magic. Run the cell below to view summary stats for all the table columns in the pedigree table of the 1000 Genomes dataset. This example may take up to one minute to run, as it is performing a query per column. Queries on public datasets are free up to 1 TB per month.

Querying with cell magics

Run the cell below to total the number of distinct families represented in the 1000 Genomes dataset.

Run the cell below to total the number of distinct families corresponding to each of the 25 populations represented in the 1000 Genomes dataset.

Run the next two cells to select all individuals for whom the dataset contains both maternal and paternal IDs, then create a plot showing the distribution of these individuals by population.

Using pandas-gbq

The `pandas-gbq` library provides a simple interface for running queries and uploading pandas dataframes to BigQuery. Additionally, this library offers utilies for creating plots from your data.

Run the cell below to use the `pandas-gbq` library to read data from BigQuery and create a plot of the data.

Provenance

Generate information about this notebook environment and the packages installed.

Conda and pip installed packages:

JupyterLab extensions:

Number of cores:

Memory:

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